

Instructions

Input Cells: Cells that require user input have an orange background.

Status Indicators: Cells, sections, and worksheets each have an overall status, indicated by blue background cells.

Images and Schematics: For sections requiring images or schematics, paste the image over the specified area. Use high-resolution images to allow for zooming if needed, and then resize the image appropriately.

Datasheets: Provide a hyperlink to the manufacturer's datasheet where requested.

Additional Comments: Sections labeled "Additional Comments" are provided for documenting anything the team feels is not adequately covered in the specified fields. Completing these sections is optional.

Status

Datasheets	OK
TS Schematics	OK
Accumulator	OK
Precharge/Discharge	OK
Charging	OK
Shutdown Circuit	BLANK
Torque Security	OK
Other	BLANK

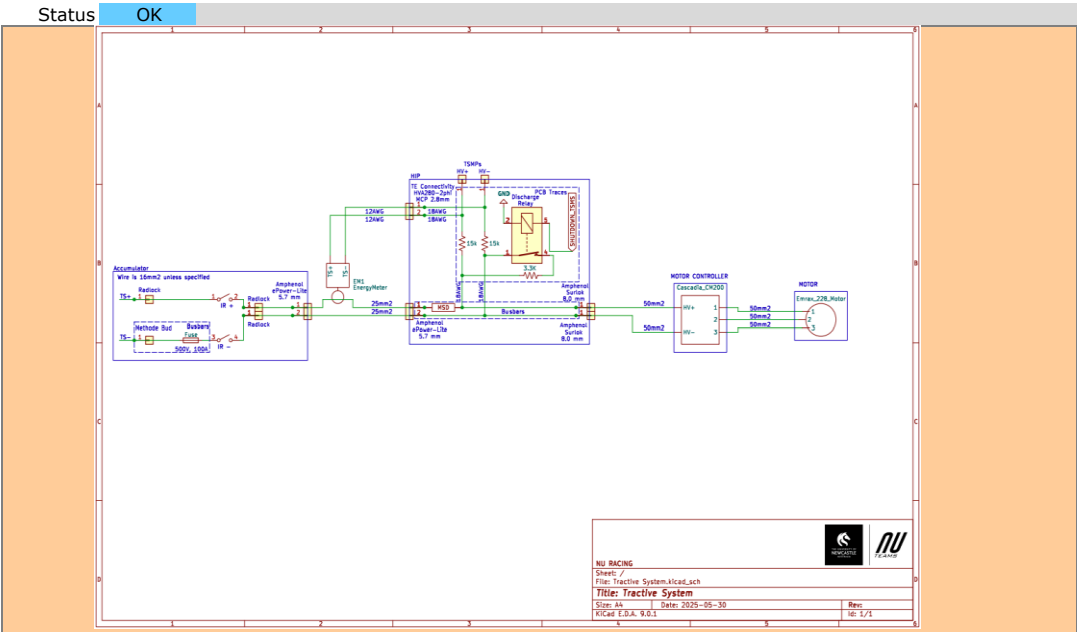
Team Name	NU Racing
University	University of Newcastle
Competition Year	2025

Max TS Voltage	453.6	V
Accumulator Fuse	100	A

Additional Comments

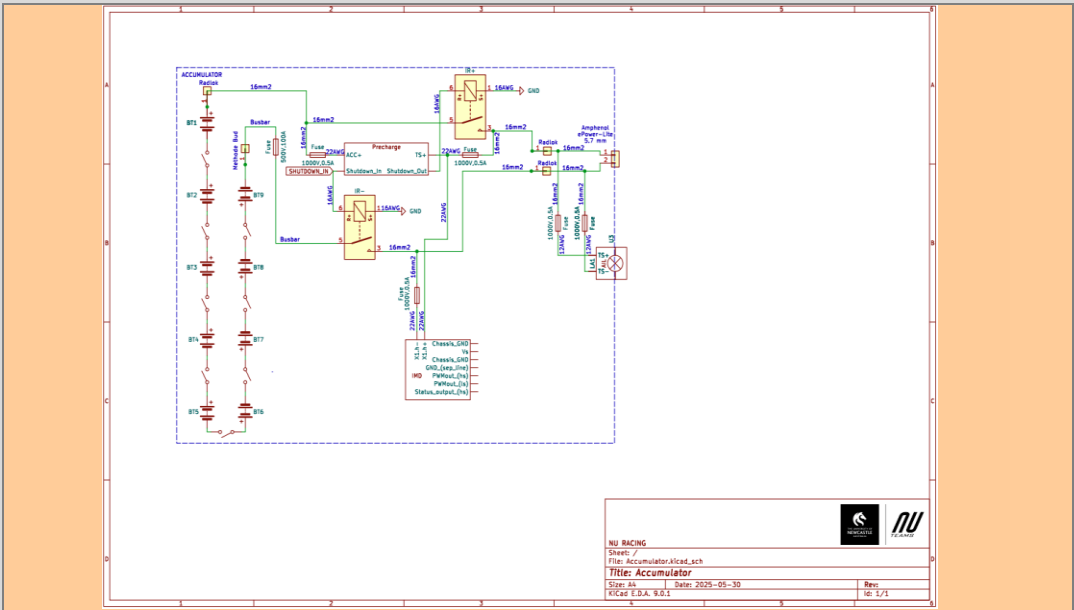
Template Revision History		
Version	Date	Comment
2.4	9/30/24	Updated for 2025 rules and terminology, general bug fixes.
2.3	10/29/23	Updated for 2024 season.
2.2	11/3/22	Updated for 2023 season.
2.1	10/22/21	Updated for 2022 season.
2.0	11/27/20	Multiple updates and bug fixes.

Contactors/Relays								OK		
Manufacturer	Part Number	Voltage [V]	Continuous Current [A]	Max Interrupt Current [A]	Normal State	Datasheet	Location Used	Additional Comments	Rules References	
GigaVac	EPICGX	800	225	500	NO	https://www.mouser.com	Accumulator	Isolation Relays	EV.5.2.1	
Omron	G7L-2A-X-	1000	20	20	NO	https://xonstorage.blob.core.windows.net	Accumulator	Precharge relay	EV.5.4	
Cynergy	DBT71210	7000	2	3	NC	https://www.cynergy.com	HIP - Discharge	Discharge relay	EV.5.6	
Connectors							OK			
Manufacturer	Part Number	Voltage [V]	Ampacity [A]	Wire Size [mm ²]	Datasheet	Location Used	Additional Comments	Rules References		
Amphenol	ELPA2A250	850	120	25	https://www.amphenol.com	Accumulator	HV loom from HIP to Accumulator, plug side	EV.5.2.1		
Amphenol	ELRA2A030	850	120	25	https://www.amphenol.com	Accumulator	Located on the HIP and Accumulator lid, receptacle bus	EV.5.9		
Methode	EBC 5.7mm	600	320	60	https://www.methode.com	Accumulator	Methode bud used on negative HV connection in accumulator	EV.6.3.3		
Amphenol Tech	RL00571-1	1000	120	25	https://au.mouser.com	Accumulator	Inline radlok connectors, used on the positive HV connection			
Amphenol	SLPPB50B5	1000	250	50	https://amphenol-ind.com	HIP - Motor	HV loom from HIP to Motor Controller, plug side, 90 degree			
Amphenol	SLPPB50B5	1000	250	50	https://amphenol-ind.com	HIP - Motor	HV loom from HIP to Motor Controller, plug side, 270 degree			
Amphenol	SLPIRBBP5	1000	250	50	https://amphenol-ind.com	HIP Enclosure	Located on HIP, receptacle bus bar, 90 degree keyway			
Amphenol	SLPIRBBP5	1000	250	50	https://amphenol-ind.com	HIP Enclosure	Located on HIP, receptacle bus bar, 270 degree keyway			
Te Connectivity	4-2103177	600	27	12	https://www.te.com/	HIP - Energy	Plug, connects DCDC and HIP HV connection. Current limit			
Te Connectivity	2103124-4	600	30	12	https://www.te.com/	HIP Enclosure	Receptacle, connects DCDC and HIP HV connection. Current			
Hirose Electric	EM30MSD0	1500	200	40	https://www.hirose.com	MSD	MSD			
Insulating Materials							OK			
Manufacturer	Part Number	Thickness [mm]	Dielectric strength [V/mm]	Voltage Rating [V]	Max Temp [C]	Flammability Rating	Datasheet	Location Used	Additional Comments	Rules References
ePOL	PEEK-NAT/	10	24000	240000	340	UL94 V-0	https://d347awuzx0k	Accumulator	Accumulator top plate material	EV.5.2.1
RS PRO	20305179	1	31500	31500	130	UL94 V-0	https://docs.rs-online.com	Accumulator	Accumulator wall lining material	EV.5.2.2.a
DuPont	Nomex 410	0.25	33000	8250	260	UL94 V-0	https://www.swiftsupplies.com	Accumulator	Accumulator lid lining material	EV.5.2.3
PCB	KB6165F T	1.6	25000	40000	170	UL94 V-1	https://odakpcb.com/	Accumulator	Accumulator cell segment lining and cell PCB material	EV.5.4.4
										EV.6.2
										EV.6.5.5.a



Vehicle Tractive System Schematic Shows			OK
Rules References			
EV.6.6.6	Shows details of all TS circuits outside of accumulator	TRUE	OK
	Accumulator is shown as a single element (without internal details)	TRUE	OK
	All wire gauges labeled	TRUE	OK
	All fuses labeled with ampacity	TRUE	OK
	All wires > 150mm have a fuse within the first 150mm from the source	TRUE	OK
	Fuse locations represent physical location	TRUE	OK
	Enclosures shown	TRUE	OK
	Connectors labeled with Make/Model	TRUE	OK
	Standard schematic symbols used	TRUE	OK
	All text is readable (zooming is allowed)	TRUE	OK
EV.5.8	TSMPs	TRUE	OK
EV.5.8.5	Motor Controller	TRUE	OK
	Motor	TRUE	OK
	TSMPs not fused	TRUE	OK

Must be shown on at least one schematic			OK
Rules References			
EV.7.6	IMD	TRUE	OK
EV.5.5	MSD	TRUE	OK
EV.5.6	Precharge	TRUE	OK
EV.5.6	Discharge	TRUE	OK
EV.3.2; EV.5.1.3	Energy Meter (shunt and TS sense connection)	TRUE	OK
EV.5.6.3.b	Discharge still functions when MSD is opened	TRUE	OK
EV.5.6.3.c	Discharge is not fused	TRUE	OK



Accumulator TS Schematic Shows			OK
Rules References			
EV.6.6.6	Details of all TS circuits within accumulator	TRUE	OK
	All wire gauges labeled	TRUE	OK
	All fuses labeled with ampacity	TRUE	OK
	All wires > 150mm have a fuse within the first 150mm from the source	TRUE	OK
	Fuse locations represent physical location	TRUE	OK
	Connectors labeled with Make/Model	TRUE	OK
	Standard schematic symbols used	TRUE	OK
	All text is readable (zooming is allowed)	TRUE	OK
	IRs	TRUE	OK
	Maintenance plugs	TRUE	OK
EV.5.4	Cells (details of all intercell connections may be abstracted)	TRUE	OK
EV.5.7	Accumulator Indicator (must be powered directly by TS)	TRUE	OK

Additional Comments

Each of the 9 accumulator segments are comprised of 12 li1x6pvtc6t packs arranged in series. These li1x6pvtc6t packs consist of 6 individual Sony Murata 18650 cells arranged in parallel.

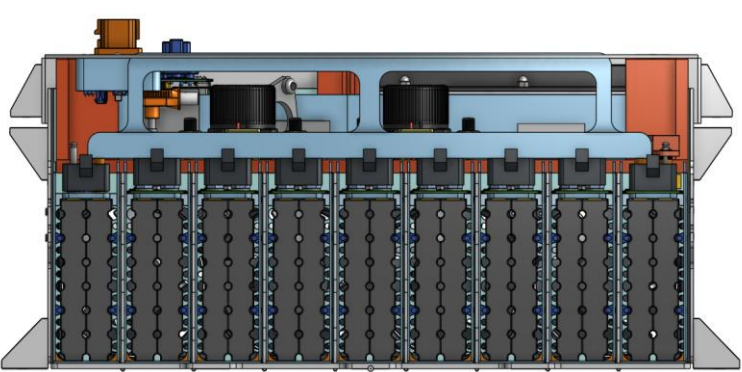
Status	OK		
Segment-to-Segment Connection	Series		OK
	Maximum TS Voltage	453.6 V	OK
	Nominal TS Voltage	388.8 V	
	Total Cells	648	

Segment Data							OK
Cell Connection Order Parallel then Series							OK
Segment	Parallel cells	Series cells	Temp Sensors	Accumulator Number	Max Voltage [V]	Segment Energy [MJ]	
1	6	12	20	1	50.4	3.374784	OK
2	6	12	20	1	50.4	3.374784	OK
3	6	12	20	1	50.4	3.374784	OK
4	6	12	20	1	50.4	3.374784	OK
5	6	12	20	1	50.4	3.374784	OK
6	6	12	20	1	50.4	3.374784	OK
7	6	12	20	1	50.4	3.374784	OK
8	6	12	20	1	50.4	3.374784	OK
9	6	12	20	1	50.4	3.374784	OK
10					0	0	
11					0	0	
12					0	0	
13					0	0	
14					0	0	
15					0	0	
16					0	0	
17					0	0	
18					0	0	

BMS			OK
BMS Type	Centralized		OK
BMS Source	Purchased		OK
Make	Ewert Energy	Model	Orion 2
Galvanic Isolation TS to GLV	1500 V		OK
Galvanic Isolation between segments	2500 V		OK
Voltage Sense Leads			
Maximum Unfused Lead Length	16 mm		OK
Sense Lead Overcurrent Protection Device	Bel Fuse C1Q 1		OK
Overcurrent Location	Other		OK
Overcurrent Other Location	CANaMon		OK
Overcurrent Voltage Rating	63 V		OK
Temperature Sensors			
Temp Sensor Location	On cell terminal		OK
Cells sensed per sensor	2 cells		OK
Thermal Contact material used			
Thermal contact material datasheet			
Thermal contact material conductivity		W/(m·K)	
Thermal contact material thickness		mm	
Percent of cells sensed	55.5556 %		OK

Maintenance Plugs			OK
Connector	Methode EBC 5.7mm		OK
Ampacity	320 A		OK
IR			
Make/Model	GigaVac EPICGX		OK
Contact Voltage Rating	800 V		OK
Max switching current	500 A		
Max continuous current	225 A		OK
Main Fuse			
Main Fuse	Littlefuse L50QS100.T		OK
Main fuse ampacity	100 A		
Main fuse voltage	500 V		OK
MSD			
MSD	Electric Group EM30M		OK
Ampacity	200 A		OK

Insulating Materials			
Between cells and accum. container	RS PRO 20305179		
Voltage Rating	31500 V		
Temperature Rating	130 C		
Flammability Rating	UL94 V-0		
On top of cells	PCB KB6165F TG 149		
Voltage Rating	40000 V		
Temperature Rating	170 C		
Flammability Rating	UL94 V-1		
Separating IRs and Fuse from cells	ePOL PEEK-NAT/BLK		
Voltage Rating	240000 V		
Temperature Rating	340 C		
Flammability Rating	UL94 V-0		

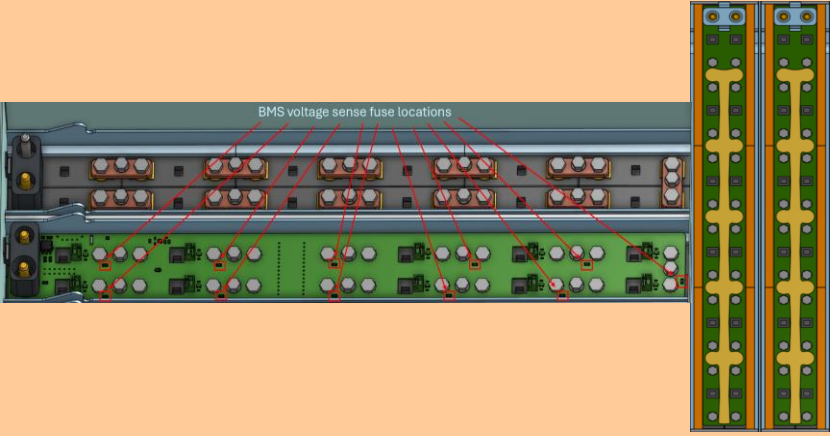
		
Maintenance Plug(s) Picture Shows		
Unique Configuration of plugs	TRUE	OK
Non-conductive on un-necessary surface	TRUE	OK
Accessible without any additional accumulator disassembly	TRUE	OK
Positive Locking	TRUE	OK
Additional Comments		Rules References
Busbar positively locked by accumulator lid.		EV.5.3

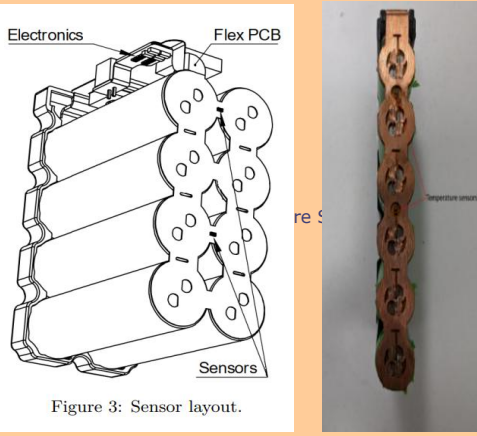
Parallel Cell Fusing			Rules Referenes
Source	Purchased	OK	EV.6.6.3
Make/Model	Energus Internal	OK	EV.6.6.5
Continuous Current	270 A	OK	

Test Data Summary		
Tests at current 1	0	
Tests at current 2	0	
Tests at current 3	0	
Tests at current 4	0	

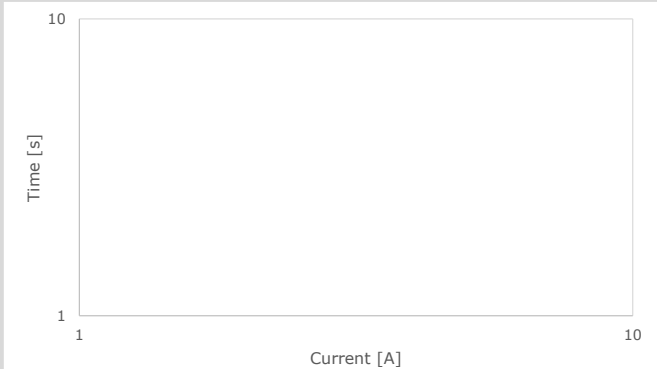
See Fusible Link Testing Guideline on FSAEOnline.com

Testing
Test 1
Test 2
Test 3
Test 4
Test 5
Test 6
Test 7

		
Cell Connections Picture Shows		
Bolted connections have positive locking	TRUE	OK
Location of parallel cell overcurrent for parallel cells	TRUE	OK
BMS voltage sense fuse location	TRUE	OK
Additional Comments		Rules References
		EV.6.4.1 EV.6.4.3 EV.7.4.4.a

		
Temperature Sensor Picture Shows		
Temp sensor in direct contact with terminal or busbar	TRUE	OK
Distance from sensor to cell negative terminal connection	TRUE	OK
Additional Comments		Rules References
Please note that the li1x6pvtc6t packs used in our accumulator are configured in a 1x6 pack, not a 4x2 as pictured above. This image was taken from the pack's data sheet which covers a range of battery configurations. The locations of the sensors is the same for both		EV.7.5

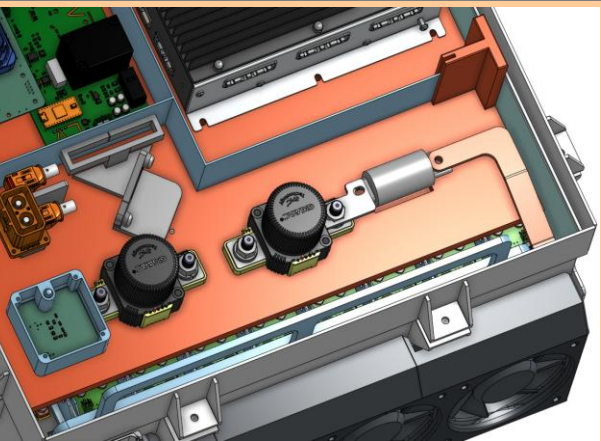
		
Fuse		



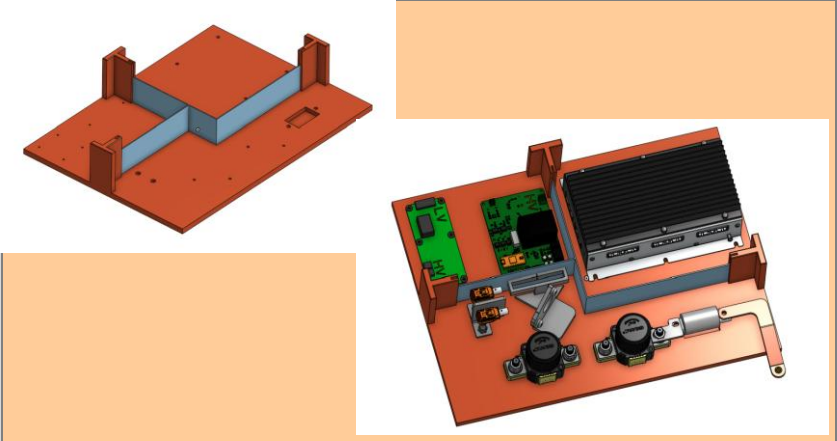
Test 8
Test 9
Test 10
Test 11
Test 12
Test 13
Test 14
Test 15
Test 16

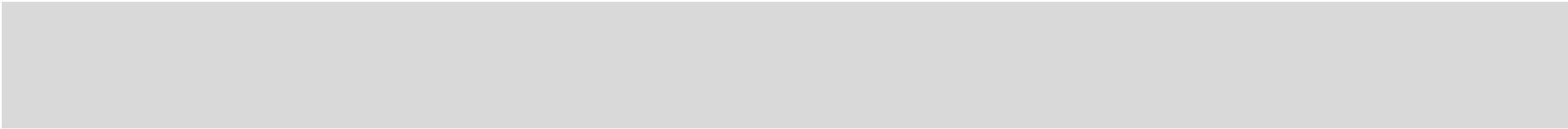


OK	Additional Comments Due to design of temperature monitoring PCB the centre 2 battery packs in each segment will not have temperature monitored.
OK	
OK	
OK	
OK	
OK	
OK	
OK	
OK	
OK	

		
and IRs Separation Picture Shows		OK
Fuse and IRs are separated from cells	TRUE	OK
Additional Comments		Rules References
		EV.5.4.4

		
Segment Insulation Picture Shows		OK
Insulation between cells and accumulator container	TRUE	OK
Additional Comments		Rules References
		EV.5.2.2.a EV.5.2.3

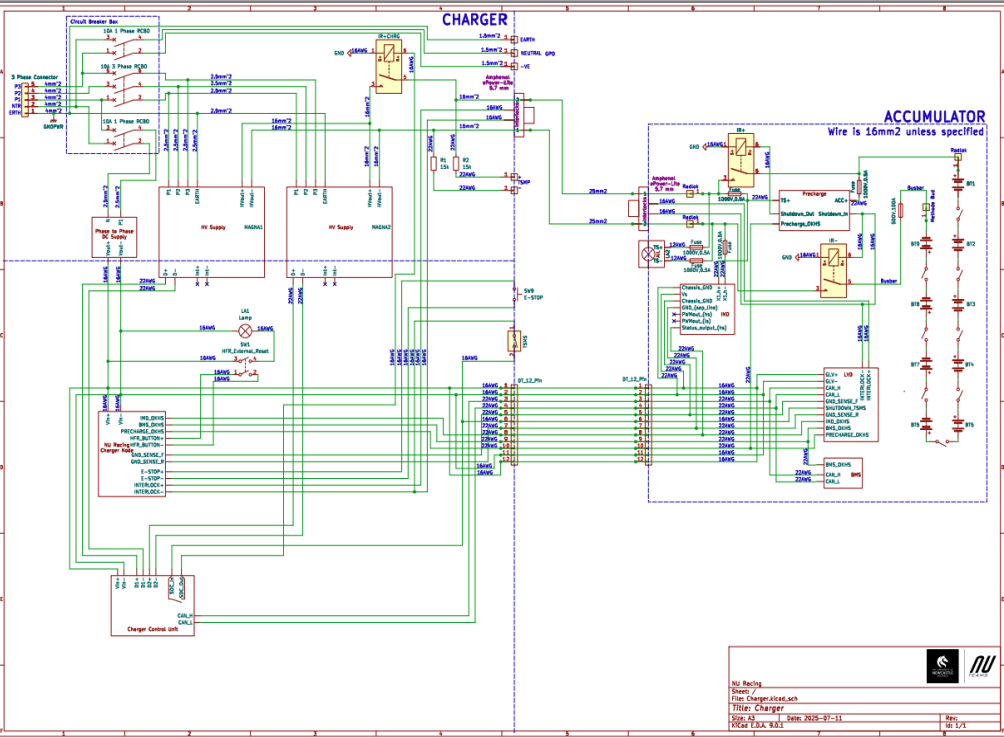
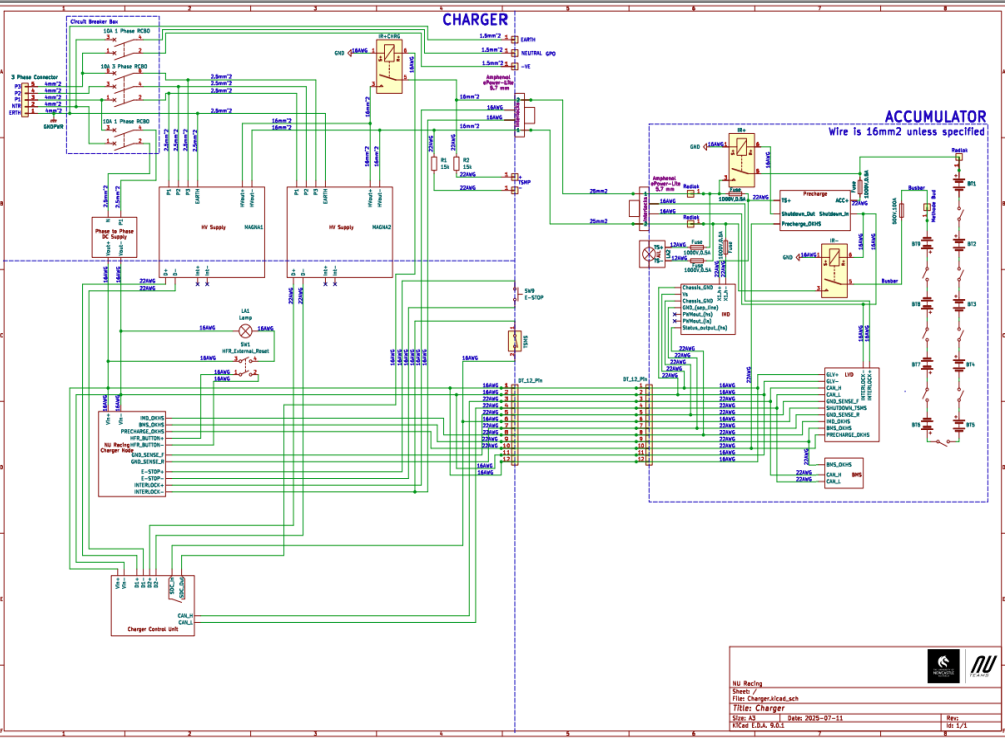
		
Segment Insulation Picture Shows		OK
Insulation on top of cells (air not acceptable)	TRUE	OK
Additional Comments		Rules References
		EV.5.2.2.a EV.5.2.3



Status OK

Charger			OK
Make	Magna Power		OK
Model	SL series		OK
Datasheet	https://magna-power.com		OK
Input Voltage	380/400	V	OK
Input Current	11	A	OK
Output Voltage	500	V	OK
Output Current	10	A	OK
Power	4	kW	OK
Input to Output isolation	2.5		OK

Additional Comments



Charging Tractive System Schematic Shows

Rules References

EV.6.6	Fuse	TRUE	OK
	Connectors	TRUE	OK
	Wire Gauge	TRUE	OK
EV.8.4	IMD	TRUE	OK
EV.8.4.2	IRs	TRUE	OK
EV.8.2.6	TSMPs	TRUE	OK

Additional Comments

Charging Shutdown Circuit Shows

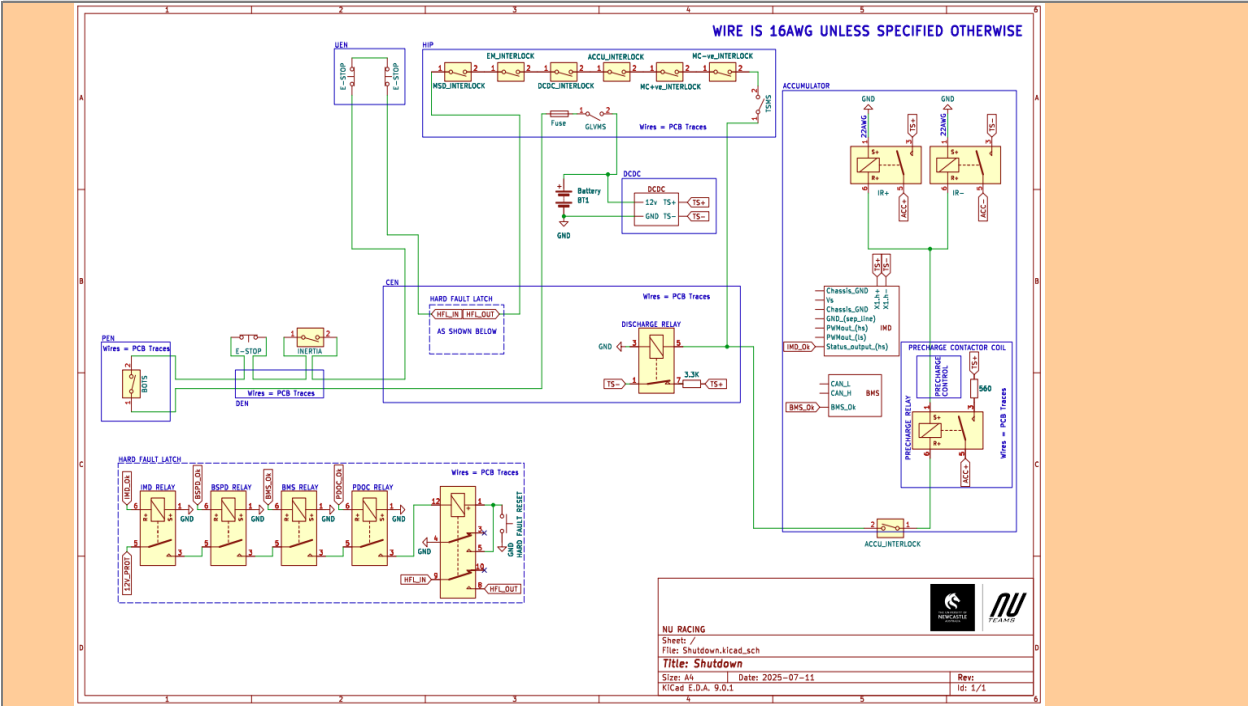
Rules References

EV.8.3	IMD Control and Powerstage	TRUE	OK
EV.8.3	AMS	TRUE	OK
EV.8.2.7	Charger Shutdown Button	TRUE	OK
EV.8.4	IRs	TRUE	OK
EV.5.9	Charge Connector Interlock	TRUE	OK
EV.8.2.4; EV.8.4.2.c	Charge Control (CAN, Enable, etc.)	TRUE	OK

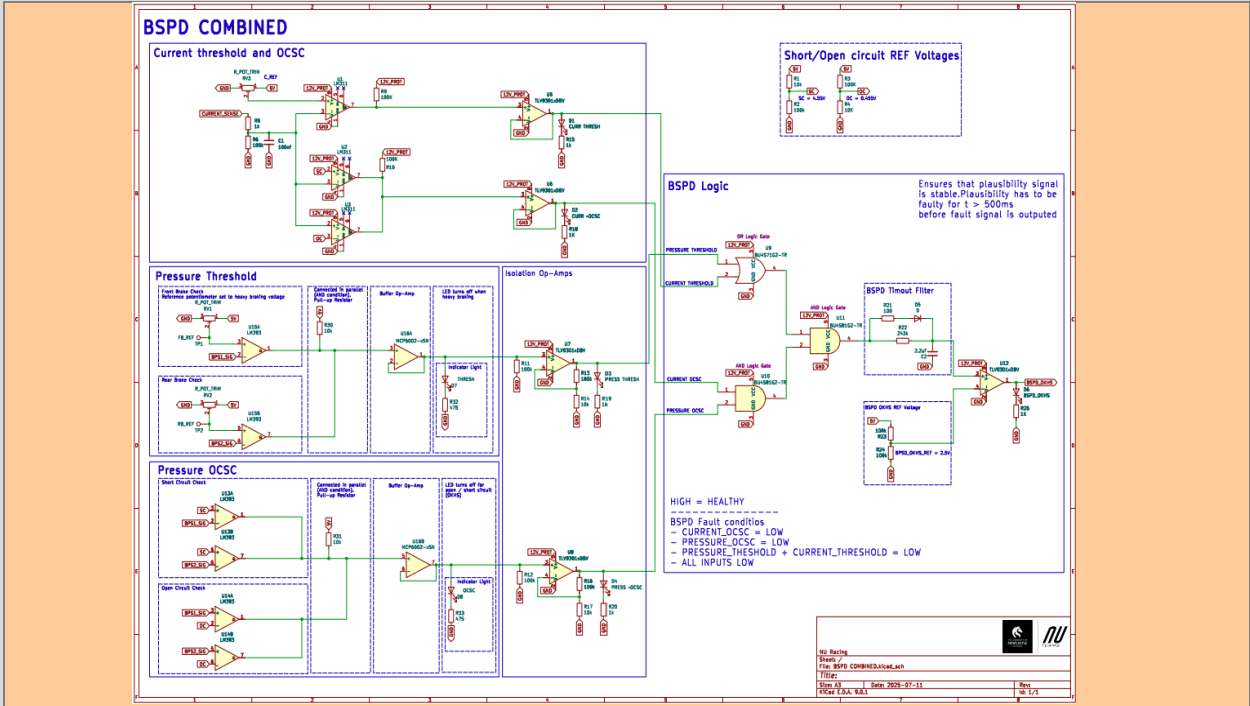
Additional Comments

Status	BLANK	
IMD		
Rules Reference	EV.7.6	OK
Make/Model	Bender IR155-3204	OK
Setpoint	250 kΩ	OK

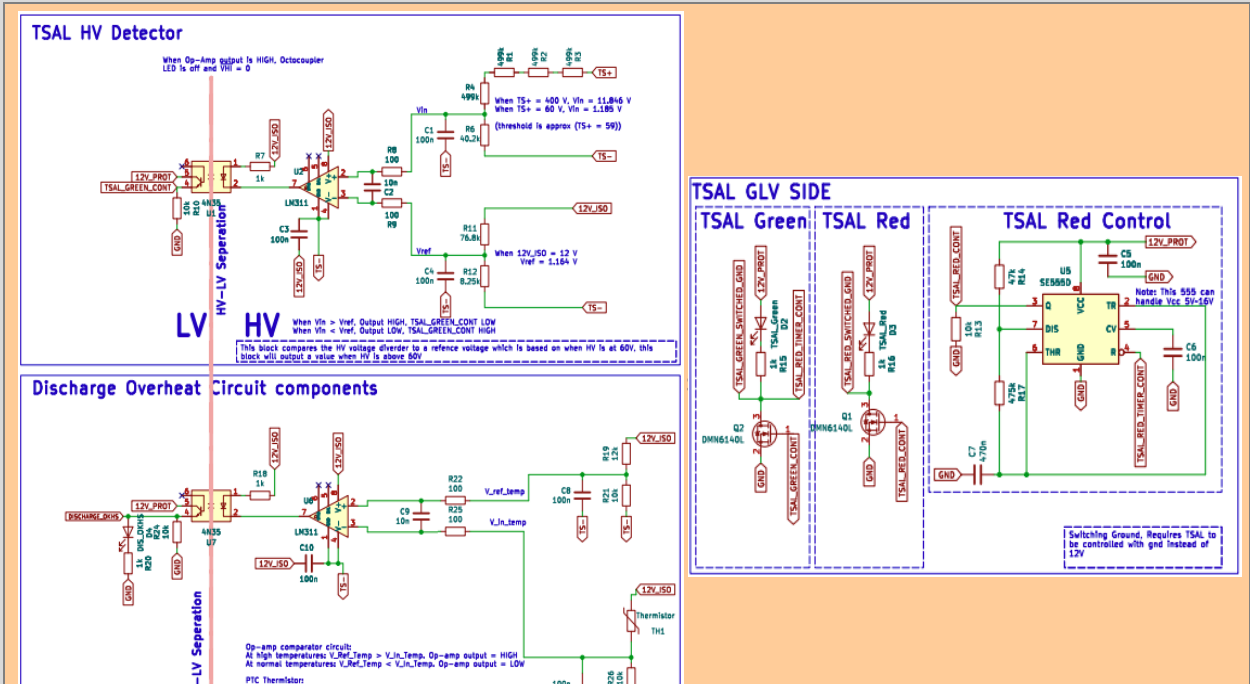
BSPD		
Rules Reference	EV.7.7	OK
Current Sensor Make/Model	Tamura L31S100S05	OK
Current Sensor Datasheet	https://au.mouser.com	OK
Trip Current	12.5 A	OK
Sensor Output @ Trip Current	2.57 V	OK
Accuracy @ Trip Current	0.1 A	OK



Shutdown Circuit Schematic Shows		OK
Rules References		
EV.7.9.2	GLVMS	TRUE OK
EV.7.9.3	TSMS	TRUE OK
EV.7.10.3	2 Side Shutdown Buttons	TRUE OK
EV.7.10.4	1 Cockpit Shutdown Button	TRUE OK
T.3.3	Brake Over Travel Switch (BOTS)	TRUE OK
T.9.4	Inertia Switch	TRUE OK
EV.7.3	BMS	TRUE OK
EV.7.6	IMD	TRUE OK
EV.7.7	BSPD	TRUE OK
EV.5.5.3	MSD Interlock	TRUE OK
EV.7.1.4	IMD, AMS, BSPD have independent power stages	TRUE OK
EV.4.1.3.a	Outboard Wheel Motor Interlocks (if applicable)	TRUE OK
	GLV Battery	TRUE OK
	HV DC/DC converter GLV connection (if applicable)	TRUE OK
	GLV Fuse	TRUE OK
	Wire Gauge	TRUE OK
	Precharge Contactor coil	TRUE OK
	Positive IR coil	TRUE OK
	Negative IR coil	TRUE OK
	Discharge Contactor coil	TRUE OK
	TSMS is last switch	TRUE OK
	GLVMS is first switch	TRUE OK
	Uses proper electrical schematic symbols	TRUE OK
	Labels/text are readable (zooming in is acceptable)	TRUE OK
EV.5.9	Interlocks on TS connectors outside of an enclosure	TRUE OK
Additional Comments		Rules References
		EV.7.1
		EV.7.2

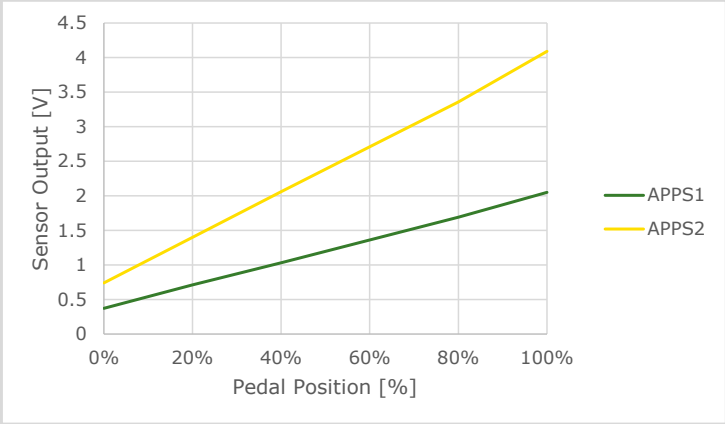


BSPD System Schematic Shows		OK
BSPD Current Sensor input open/short circuit protection		TRUE OK
BSPD Brake Position input open/short circuit protection		TRUE OK
Shutdown circuit powerstage		TRUE OK
Additional Comments		Rules References
		EV.7.7



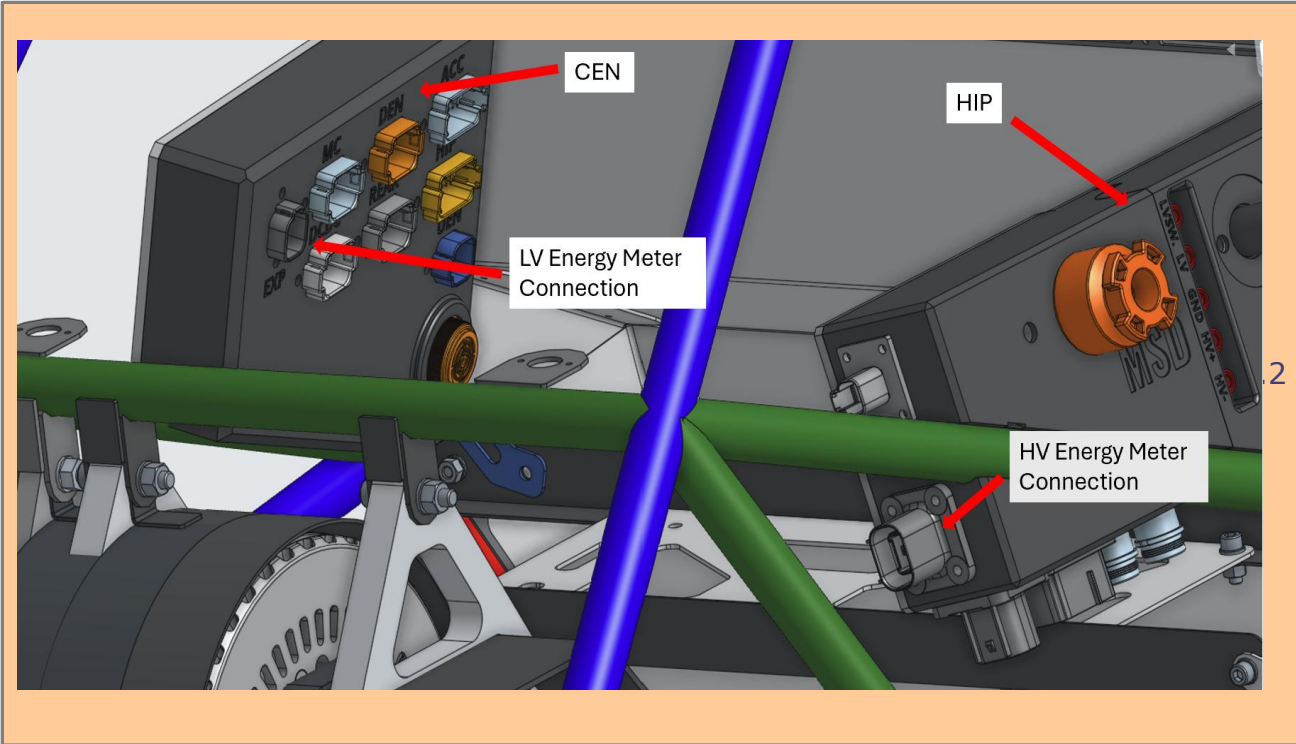
	LV HV When the temperature increases, resistance increases. Model: TFP1206L1003FM 10k ohm nominal, 12.5k ohm @ 80 C 100n C11 10k R1			
	Ready to Move Light Schematic Shows			BLANK OK OK OK OK
	Light controlled by >60V present in TS (hardware-based voltage detection)		TRUE	
	Light powered by GLV		TRUE	
	Light flashes at 2-5 Hz when >60V present in TS		TRUE	
	Light Source		Team Built	
	DOT FMVSS 108 compliant			BLANK
	Additional Comments			Rules References
				EV.5.10

Status	OK		
APPS Sensor Transfer function			OK
Rules Reference	T.4.2.3		
Position	APPS1 Output [V]	APPS2 Output [V]	
0%	0.372	0.74	OK
20%	0.711	1.4	OK
40%	1.03	2.06	OK
60%	1.36	2.71	OK
80%	1.69	3.36	OK
100%	2.05	4.09	OK

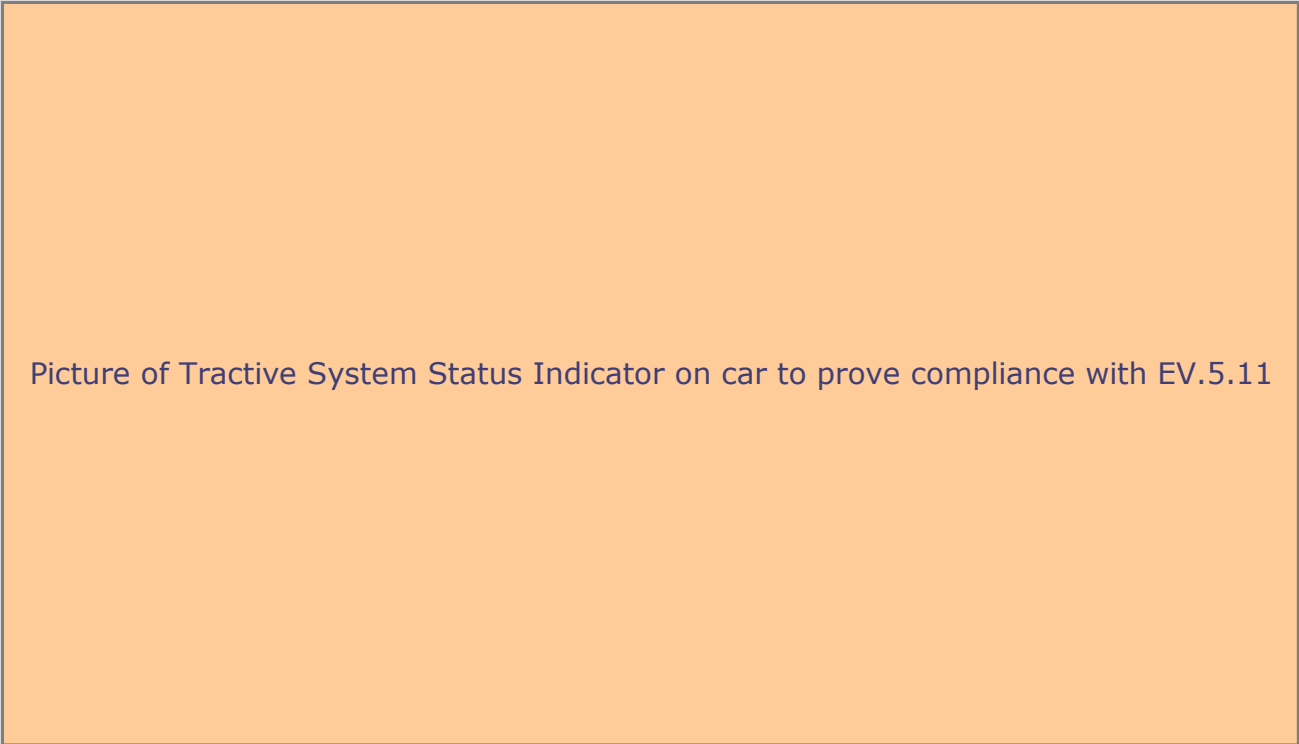


Torque Control Path Security Checks									OK	
Out Of										
Source Device	Destination Device	Communication	Redundant	Range	Correlation	Checksum	Timeout	Other		Additional Comments
APPS	Pedal Box ECU	Analog	2 Sensors	Yes	No	No	No		OK	
Pedal Box ECU	DEN	Digital	No	No	No	CAN	Yes		OK	
DEN	CEN	Digital	No	No	No	CAN	Yes		OK	
CEN	Motor Controller	Digital	No	No	No	CAN	Yes		OK	
Motor Controller										

Status	BLANK					
Motor Controller			OK	Motor		OK
Make	Cascadia		OK	Make	Emrax	OK
Model	CM200DX		OK	Model	228 MV	OK
Datasheet	https://www.cascadia.com		OK	Datasheet	https://emrax.com/w	OK
Galvanic Isolation between TS and control	1500	V	OK			



Energy Meter Download Picture Shows		OK
Energy Meter Download Connector	TRUE	OK
Additional Comments		



Tractive System Status Indicator Shows		BLANK
Tractive System Status Indicator placement in vehicle		BLANK
Additional Comments		